

ANSIBLE

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LAB 1: WORKING WITH BASIC ADHOC COMMANDS IN ANSIBLE

- We will see how we can create a file, install a package, create a directory , remove a package with a single line command(Adhoc)

In Ansible we have Ansible modules which is a small unit of work that ansible executes on the target machine .

Ansible modules are small , reusable units of code that ansible uses to perform specific task on the managed nodes .They are like building blocks of ansible

Lab 1.1 Create a blank file using ansible adhoc module

- https://docs.ansible.com/projects/ansible/latest/collections/ansible/builtin/file_module.html

Scroll down till examples

```
- name: Touch a file, using symbolic modes to set the permissions (equivalent to 0644)
  ansible.builtin.file:
    path: /etc/foo.conf
    state: touch
    mode: u=rw,g=r,o=r
```

Ansible module
(you can also write it as file)

File will be created in which path. In this example blank
file with name foo.conf is created

We need to create blank file

ansible all -m file -b -a "path=/home/devops/akshat.txt state=touch"

all : all the nodes . You can put the group name as well here

-b: become : This task will be executed with sudo permissions

-m : module.

-a : argument

```
devops@ip-172-31-71-170:~$ ansible all -m file -b -a"path=/home/devops/akshat.txt state=touch"
[WARNING]: Host '172.31.72.173' is using the discovered Python interpreter at '/usr/bin/python3.12', but
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/referen
information.
172.31.72.173 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": true,
  "dest": "/home/devops/akshat.txt",
  "gid": 0,
  "group": "root",
  "mode": "0644",
  "owner": "root",
  "size": 0,
  "state": "file",
  "uid": 0
}
[WARNING]: Host '172.31.68.157' is using the discovered Python interpreter at '/usr/bin/python3.12', but
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/referen
information.
172.31.68.157 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": true,
  "dest": "/home/devops/akshat.txt",
  "gid": 0,
  "group": "root",
  "mode": "0644",
  "owner": "root",
  "size": 0,
```

Lab 1.2 : Use the command module and run linux command

- List the files

```
devops@ip-172-31-71-170:~$ ansible all -m command -b -a "ls"
[WARNING]: Host '172.31.72.173' is using the discovered Python interpreter at '/usr/bin/python3' which
could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible/latest/reference_appendices/config.html#interpreter_python
172.31.72.173 | CHANGED | rc=0 >>
akshat.txt
[WARNING]: Host '172.31.68.157' is using the discovered Python interpreter at '/usr/bin/python3' which
could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible/latest/reference_appendices/config.html#interpreter_python
172.31.68.157 | CHANGED | rc=0 >>
akshat.txt
devops@ip-172-31-71-170:~$
```

- Create a blank file using command module

```
devops@ip-172-31-71-170:~$ ansible all -m command -b -a "touch file1"
[WARNING]: Host '172.31.68.157' is using the discovered Python interpreter at '/usr/bin/python3' which
could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible/latest/reference_appendices/config.html#interpreter_python
172.31.68.157 | CHANGED | rc=0 >>

[WARNING]: Host '172.31.72.173' is using the discovered Python interpreter at '/usr/bin/python3' which
could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible/latest/reference_appendices/config.html#interpreter_python
172.31.72.173 | CHANGED | rc=0 >>
```

Lab 1.3 Install apache package using adhoc module

- Go to https://docs.ansible.com/projects/ansible/latest/collections/ansible/builtin/package_module.html

Examples

```
- name: Install ntpdate
  ansible.builtin.package:
    name: ntpdate
    state: present
```

```
sense in ad-hoc (include, meta, etc)
$ ansible all -m package -b -a "name=apache2 state=present"
```

State : present : install the package

Absent: remove the package

Latest: update the package

We can confirm the apache is installed with the command module

```
devops@ip-172-31-71-170:~$ ansible all -m command -b -a"which apache2"
[WARNING]: Host '172.31.72.173' is using the discovered Python interpreter at '/usr/bin/python3' which
could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible/latest/user_guide/playbooks_warnings.html#python-interpreter-warning for more
information.
172.31.72.173 | CHANGED | rc=0 >>
/usr/sbin/apache2
[WARNING]: Host '172.31.68.157' is using the discovered Python interpreter at '/usr/bin/python3' which
could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible/latest/user_guide/playbooks_warnings.html#python-interpreter-warning for more
information.
172.31.68.157 | CHANGED | rc=0 >>
/usr/sbin/apache2
devops@ip-172-31-71-170:~$
```

Lab 1.4 Deploy hello world website in nodes using adhoc commands

Command 1: `devops@ip-172-31-71-170:~$ ansible all -m package -b -a "name=apache2 state=present"`

Command 2: `devops@ip-172-31-71-170:~$ ansible all -m command -b -a "rm /var/www/html/index.html"`

Command 3: Create a file in the ansible server with the the name index.html and content hello world

```
devops@ip-172-31-71-170:~$ ansible all -m copy -b -a "src=/home/devops/index.html dest=/var/www/html/index.html "  
[WARNING]: Host '172.31.72.173' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation of another
```

We have copied the index.html from the ansible server to the nodes

If you are asked to write these commands 100 times it will be difficult. To solve this issue we have ansible playbook

PLAYBOOKS

- Files written in YAML format which consist of the list of tasks that need to be performed in the nodes. You can have 10000000000sssss... of task in a single playbook
And when you execute the playbook all those tasks will be executed in the nodes
- Playbook follows idempotency which means if the task is already performed in the nodes it will not be reperformed if you keep running the playbook again and again

```
---
- hosts: all
  become: true
  tasks:
    - name: Install docker
      package:
        name: docker.io
        state: present
    - name: Touch again the same file
      file:
        path: /home/devops/mynewfile
        state: touch
```

Execute playbook

```
devops@ip-172-31-71-170:~$ ansible-playbook playbook.yml

PLAY [all] *****

TASK [Gathering Facts] *****
[WARNING]: Host '172.31.72.173' is using the discovered Python interpreter which
could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible/latest/reference_appendices/config.html#ansible-python-interpreter
ok: [172.31.72.173]
[WARNING]: Host '172.31.68.157' is using the discovered Python interpreter which
could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible/latest/reference_appendices/config.html#ansible-python-interpreter
ok: [172.31.68.157]

TASK [Install docker] *****
changed: [172.31.72.173]
changed: [172.31.68.157]

TASK [Touch again the same file, but do not change times this makes the] *****
changed: [172.31.72.173]
```

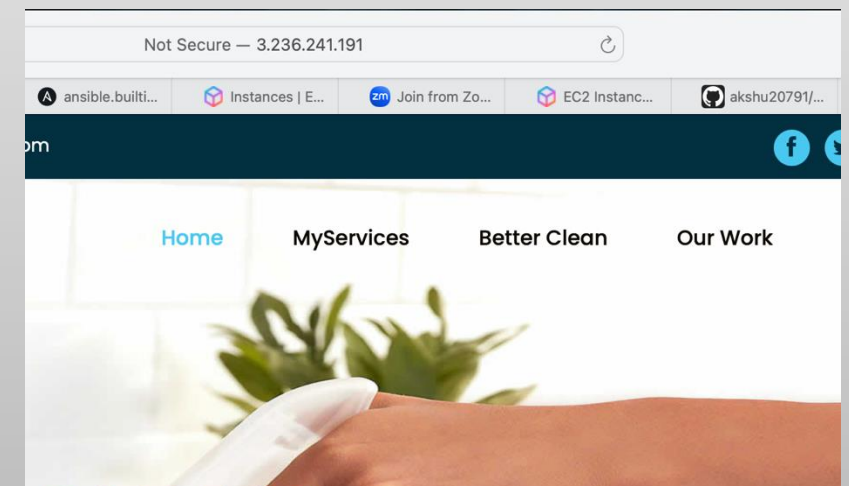
Create a playbook to deploy Apache application on the nodes

<https://github.com/akshu20791/apachewebsite>

```
---
- hosts: all
  become: true
  tasks:
    - name: Install apache
      ansible.builtin.package:
        name:
          - apache2
          - git
        state: present
    - name: Remove file (delete file)
      ansible.builtin.file:
        path: /var/www/html/
        state: absent
    - name: Git checkout on apache default location
      ansible.builtin.git:
        repo: 'https://github.com/akshu20791/apachewebsite'
        dest: /var/www/html/
```

```
devops@ip-172-31-71-170:~$ ansible-playbook playbook2.yml
PLAY [all] *****
```

Open public ip of node 1 or node 2



But if you keep running this playbook it will not follow idempotency because we are giving the instructions like delete the files ...basically this instruction should be condition based ...like for example if we install apache then only the files at /var/www/html should be removed otherwise we don't have to remove
THIS WE CALL IT AS A HANDLERS ...(conditional based tasks which will only execute if other task execute)

Ansible handlers are special tasks that execute only when explicitly notified by another task within a playbook or role. They are primarily used for reacting to changes in the system's state or configuration, ensuring that actions are performed only when necessary.

Supporting doc:

https://docs.ansible.com/projects/ansible/latest/playbook_guide/playbooks_handlers.html

Here if apache and git is installed then only we want to remove /var/www/html and do git checkout so we will use handlers...where notify will only be executed if install apache and git task is executed

```
---
- hosts: all
  become: true
  tasks:
    - name: Install apache and git
      ansible.builtin.package:
        name:
          - apache2
          - git
        state: present
      notify: Remove file
      notify: Git checkout on apache default location
  handlers:
    - name: Remove file
      ansible.builtin.file:
        path: /var/www/html/
        state: absent
    - name: Git checkout on apache default location
      ansible.builtin.git:
        repo: 'https://github.com/akshu20791/apachewebsite'
        dest: /var/www/html/
```

Variables

A variable is a name that stores a value. Instead of writing the same value again and again you store it in a variable and use it everywhere

- ✓ Makes playbook clean
- ✓ Makes playbook reuseable
- ✓ Makes playbook Dynamic

```
---
- hosts: all
  become: true
  vars:
    my_dest: /home/devops/akshatfile.txt
    dir_loc: /home/devops/mydir
  tasks:
    - name: Touch a file, using symbolic modes to set the permissions (equivalent to 0644)
      ansible.builtin.file:
        path: "{{ my_dest }}"
        state: touch
    - name: Create a directory if it does not exist
      ansible.builtin.file:
        path: "{{ dir_loc }}"
        state: directory
```

In this example we are creating two variables my_dest and other is dir_loc Now , we are passing the values of these variables in the tasks

- Variable in a separate file:

vi vars.yml

```
my_dest: /home/devops/mynewnewfile
dir_loc: /home/devops/mynewdir
```

Here we are passing the variable in a different file known as vars.yml

And we will call that file location in the playbook

Create a playbook vi mynewplaybook.yml

```
---
- hosts: all
  become: true
  vars_files:
    - vars.yml
  tasks:
    - name: Touch a file, using symbolic modes to set the permissions (equivalent to 0644)
      ansible.builtin.file:
        path: "{{ my_dest }}"
        state: touch
    - name: Create a directory if it does not exist
      ansible.builtin.file:
        path: "{{ dir_loc }}"
        state: directory
```

```
devops@ip-172-31-71-170:~$ vi playbook3.yml
devops@ip-172-31-71-170:~$ ansible-playbook playbook3.yml
```

```
PLAY [all] *****
TASK [Gathering Facts] *****
```

- Variables as a prompt

```
---
- hosts: all
  become: true
  vars_prompt:
    - name: my_dest
      prompt: Put the file location where you ned to create file
      private: false
    - name: dir_loc
      prompt: put the directory location
      private: false
  tasks:
    - name: Touch a file, using symbolic modes to set the permissions (equivalent to 0644)
      ansible.builtin.file:
        path: "{{ my_dest }}"
        state: touch
    - name: Create a directory if it does not exist
      ansible.builtin.file:
        path: "{{ dir_loc }}"
        state: directory
~
```

When you execute the playbook it will prompt you for the values

```
devops@ip-172-31-71-170:~$ ansible-playbook playbook3.yml
Put the file location where you ned to create file: /home/devops/myakshatfie
put the directory location: /home/devops/akshatdir

PLAY [all] *****

TASK [Gathering Facts] *****
[WARNING]: Host '172.31.72.173' is using the discovered Python interpreter at '/usr/bin/
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansibl
information.
```

When you have list of values that need to be passed as a package then use pkg_list

```
---
- hosts: all
  become: true
  vars:
    pkg_list:
      - apache2
      - git
  tasks:
    - name: Install apache and git
      ansible.builtin.package:
        name: "{{ pkg_list }}"
        state: present
        notify: Remove file
        notify: Git checkout on apache default location
  handlers:
    - name: Remove file
      ansible.builtin.file:
        path: /var/www/html/
        state: absent
    - name: Git checkout on apache default location
      ansible.builtin.git:
        repo: 'https://github.com/akshu20791/apachewebsite'
        dest: /var/www/html/
```

- **Ansible vault :**

Security feature used to encrypt sensitive data such as

- ☐ passwords
- ☐ SSH private keys
- ☐ Api keys
- ☐ Database credentials
- ☐ Any sensitive variable



It ensures only authorized people can view the data

- Playbook remain secure
- Even if uploaded to github, the password protected data remains safe
- Only decrypted during runtime with vault password

Click below on LABS to access the ansible vault lab

[LABS](#)

- Ansible roles :

It is a way to organize your playbook into clean, reusable structure

- i. It breaks a big playbook into small parts
- ii. Make code reusable
- iii. Best practice for production
- iv. Perfect for devops projects

```
devops@ip-172-31-71-170:~$ tree minrole
minrole
├── README.md
├── defaults
│   └── main.yml
├── files
├── handlers
│   └── main.yml
├── meta
│   └── main.yml
├── tasks
│   └── main.yml
├── templates
├── tests
│   ├── inventory
│   └── test.yml
└── vars
    └── main.yml
```

Without roles -> your playbook become HUGE

You will have 40-50 tasks , handlers, variables, templates, files

In a single file = **X** confusion

With roles -> everything is neatly arranged

Ansible galaxy

- It is public market place for roles and collections

It gives you:

- Pre built roles
- Pre built collections
- Reuseable automation content

Etc

- ANSIBLE ROLES LAB:

[CLICK HERE TO ACCESS LABS](#)